Streaming Key Estimation with Abstention from Live Chord-Recognition Output

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Abstract. Most key-detection research is retrospective: a system reads a complete score or recording and then reports its key. We study the interactive version of the problem. A musician plays a MIDI keyboard, a chord recognizer names each chord as it lands, and a key detector must decide after every chord which key the player is in, or admit that the evidence is not yet sufficient. We define an evaluation protocol for this streaming task that measures abstention, stability, and key-change lag alongside accuracy, and we evaluate a family of causal detectors on four annotated corpora. The selected system is a hidden Markov model (HMM) run strictly forward over pitch-profile evidence, with one constrained cue that separates major from minor keys sharing the same tonic. The central finding is that the detector's evidence memory selects which kind of annotated key it reports: short memory tracks the brief local keys marked by music analysts, long memory tracks the section-level keys of song annotations, and neither setting is simply more accurate, so accuracy comparisons between key detectors are incomplete without the label timescale. Most added complexity fails to help: weighting by the chord recognizer's confidence, progression rules, and adaptive changepoint models all leave the selected operating point unimproved. On held-out pop songs, the causal, abstaining detector performs at least on par with standard offline key finders that read the entire song before answering.

1 Introduction

A musician improvises at a MIDI keyboard while an application identifies each chord as it is played. A natural next feature is a key indicator: a small display answering, from the playing alone, "what key am I in right now?" Key finding is one of the oldest problems in music information retrieval, so it sounds like a solved problem waiting to be wired in. It is not: nearly all published key finding is retrospective, reading a complete score or recording and only then reporting a key, either one label for the whole piece or a local analysis computed with the entire piece in view. A live indicator enjoys no such hindsight, and its setting differs from the literature's in three compounding ways.

First, the detector is **causal and streaming**. It only ever sees the past, and it must update after each chord, from someone noodling with finger rolls, pedal blur, and wrong notes. There is no second pass and no lookahead, so the whole-sequence Viterbi decodings that dominate published HMM key finding are unavailable by construction.

Second, the observations are **uncertain and symbolic at the chord level**. The input is neither ground-truth notes nor audio chroma but the output of a chord recognizer: ranked candidate identities with explanation costs (the recognizer's own measure of how well the sounding notes fit each candidate), plus the voicing, timing, and hold duration of each committed chord. The detector therefore knows each chord's root, quality, and bass, an unusual asset, but the recognizer itself can be wrong, an unusual noise source.

Third, **abstention is a first-class outcome**. Some music hovers deliberately between keys; a two-chord modal vamp has no single right answer, and asserting one confidently is worse than staying quiet.

These constraints change what counts as a good answer. A detector can score well on conventional accuracy while flickering between keys, guessing through ambiguous passages, and trailing every key change; whole-piece evaluation notices none of this. We therefore measure coverage (how often the detector speaks) and accuracy on claimed events as an inseparable pair, along with stability, modulation lag, and time-to-first-claim. Defining the task this carefully pays off beyond the interactive setting: the load-bearing choice in the final system is the length of the detector's evidence memory, and what that dial selects is which kind of annotated "key" the detector reports, brief local tonicizations or the broader section key. Accuracy comparisons between key detectors are under-specified until the evaluation says which target counts as correct.

The paper's contributions follow that arc. We define the task and an evaluation protocol for streaming key estimation with abstention, frozen before tuning, with piece-level paired statistics for every adoption decision, a held-out evaluation performed exactly once, and a fixture-based reproducibility design (Section 3, Section 4). We establish the timescale finding between corpora, within one corpus on identical performed input, and on held-out data with significance in both directions (Section 6). We map which symbolic and temporal additions earn their keep: one adopted cue, a mass-preserving same-tonic mode tilt (Section 7), against measured negatives for recognizer-confidence weighting, progression rules, relative-pair tilts, and adaptive temporal models (Section 6, Section 7, Section 8). What these measurements leave standing is a validated, interpretable baseline: a causal HMM over profile-correlation emissions with posterior-margin abstention (Section 5) that reaches at least parity with offline whole-piece baselines on held-out data (Section 9).

2 Related work

Distributional key finding begins with the Krumhansl-Schmuckler probe-tone profiles [1] and their revisions [2], with corpus-trained profiles improving minor-mode behavior [3]. These methods estimate key from pitch-class distributions and remain important baselines because they are simple, reproducible, and musically interpretable. The selected detector in this paper keeps that interpretability: its emissions are profile correlations rather than learned sequence embeddings.

Temporal structure enters key and harmony estimation through Bayesian and hidden Markov treatments [4], [5]. The closest published analogue of our final detector is justkeydding [6], an HMM over profile emissions for global and local key. Neural symbolic-harmony systems also estimate local key as one head of a larger harmonic-analysis task, for example models using Roman numeral corpora and multitask architectures [7], [8]. Those systems demonstrate the value of local-key estimation, but they analyze complete scores offline, where our detector must filter causally over chord-recognition events and abstain when the evidence is thin.

Real-time tonal tracking has also been studied outside the profile/HMM family. Chew's Spiral Array and Center of Effect Generator frame key finding as movement in a geometric tonal space, tracking an evolving tonal center as notes accumulate [9]. Our setting differs in the observation layer: the detector receives symbolic chord identities and candidate rankings from an upstream recognizer rather than raw pitch collections.

Bayesian online changepoint detection (BOCPD) [10] tracks a posterior over how long the current regime has lasted, making it the natural adaptive-memory alternative to a fixed decay. Recent extensions model within-regime dynamics to suppress false alarms [11]. We build the constant-hazard version for the 24-key space and measure where its adaptivity helps and where it conflicts with the selected section-key operating point.

Evaluation practice follows the MIREX audio key task's weighted scoring [12] for near misses, applied to our own corpora (scores are not comparable to MIREX leaderboards, which are audio-based on different data). Ground truth spans two annotation cultures, a distinction central to Section 6. Analyst-marked local keys record every brief modulation: When in Rome's RomanText analyses [13], [14]. Section-level keys name a stable home key: Isophonics song keys via the ChoCo chord corpus [15], [16], and the key signatures of the ASAP performed piano MIDI corpus [17].

3 Task and evaluation protocol

Task. The input is a causal stream of chord events from the application's capture path: each event carries ranked chord candidates with explanation costs, the sounding voicing, the analysis-time tonality context, a timestamp, and a hold duration. An event is one held chord, not one keystroke: the capture path aggregates pedal-aware sounding notes and commits an event only after its chord identity persists past a debounce and minimum-duration gate, so finger rolls and passing voicings are absorbed upstream of the detector. After each event the detector outputs either a ranked list of (key, confidence) hypotheses or an explicit abstention. It never sees the future and is never re-run on edited history. The key space is the 24 major and minor keys over pitch classes; enharmonic spelling is a presentation concern.

Metrics. The top-ranked claim is scored. Accuracy-like metrics are reported in the selective-prediction frame [18]: coverage (how often the detector makes a claim) and accuracy on claimed events form an inseparable pair, and abstentions are never counted as errors. Exact accuracy is complemented by the MIREX-weighted score, which grants partial credit for musically close misses (fifth 0.5, relative 0.3, parallel 0.2) [12].

Temporal behavior gets three measurements. An annotated key change is **matched** when the detector claims the new key before the next change, and its **lag** is counted in events; unmatched changes are reported as censored counts, never averaged into lag. A switch between claims is **spurious** only when the annotation did not change and the new claim is not the annotated key, so a lagged catch-up onto the right key is never penalized twice. **Time-to-first-claim** counts events before the detector first commits. These counts are long-tailed, so each is summarized by the per-piece median and 90th percentile (p90) rather than a mean. Ambiguity-labeled events (hand-authored fixtures only) accept abstention or any acceptable key.

Two diagnostics complete the picture. Selective-prediction behavior is the coverage-accuracy curve swept over the abstention threshold, with the selected operating point marked. For probabilistic detectors we also report top-label posterior reliability: events are binned by the posterior probability of the leading key (ten equal-width bins) and compared with exact correctness, alongside expected calibration error (ECE), negative log likelihood, and Brier score. Reliability is separate from abstention: a detector can abstain informatively while still being overconfident in its posterior probabilities.

Statistics. Every adoption decision compares two configurations piece by piece, using the Wilcoxon signed-rank test and a seeded-bootstrap 95% confidence interval on the mean per-piece difference. The piece is the unit of analysis because events within a piece are strongly dependent and a few long pieces dominate pooled counts; pooled event accuracy is never decisive. The signed-rank test assumes no distributional form for these small samples. P-values on development data served model selection and are uncorrected for multiplicity; confirmatory weight rests on the pre-declared held-out evaluation, performed exactly once (Section 9).

Protocol discipline. The protocol was frozen before detector tuning, with changes recorded as dated amendments. Devel-

corpus	input	labels	pieces	events
When in Rome [13]	quantized scores	analyst local keys (local-key)	77	5,207
ASAP [17]	performed piano MIDI	key signatures (section-key, mode-unknown)	60	19,546
Isophonics [15], [16]	synthesized voicings	song keys (section-key, mode-resolved)	224	19,062
ASAP-WiR overlap	performed piano MIDI	analyst local keys (local-key)	36	10,395

Table 1: Evaluation corpora. Development/test splits are frozen for the first three; the overlap corpus is evaluation-only (every configuration run on it was committed in advance).

opment/test splits were frozen per corpus before the first experiment on it (by piece, and by composer where the corpus allows); all tuning, ablation, and model selection ran on development splits; the test splits were evaluated exactly once, with the result set declared in advance (Section 9). The evaluation harness structurally strips labels before events reach a detector.

4 Data and reproducibility

Evaluation uses **fixtures**: versioned event streams containing the chord recognizer's candidate rankings, the observed voicings, timing metadata, and a parallel label stream read only by the scorer. Fixtures are generated under a fixed neutral analysis context: the recognizer never sees the annotated key, so its tonality-sensitive ranking rules cannot leak ground truth into the observations. Results are comparable only within a fixture version, because fixture contents embed a specific version of the chord-recognition engine.

Four fixture sets span two annotation cultures and three input conditions (Table 1). The first three have frozen development/test splits. The ASAP-WiR overlap is evaluation-only: every configuration run against it was declared in advance because the set exists specifically to test whether the timescale effect survives on identical performed input.

The overlap corpus transfers When in Rome analyst keys onto ASAP performances of the same Beethoven sonata movements through the performance-to-score downbeat alignment, giving real tonic-and-mode ground truth on performed input. A small hand-authored pop/jazz suite (12-bar blues, modal vamps, deliberately ambiguous loops) serves as a per-fixture behavioral regression battery outside all pooled statistics. Performed corpora are replayed through the capture path of Section 3, so detectors see the events they would see live.

The repository contains the open fixtures, split files, evaluation harness, paired-comparison script, dated experiment logs, and the committed artifacts of the single held-out evaluation. License-gated corpora are not committed: their extractors refuse to write inside the research directory and must be rerun locally from pinned upstream checkouts. This means the paper's reported numbers are auditable from committed reports and commands, while redistribution remains bounded by each source corpus's terms.

5 Model family and selected detector

The detector family was restricted to components a musician could inspect (profile matches, chord-function scores, cadence votes), with no learned embeddings; the selected detector is the simplest member that survived measurement. The family's floor is profile correlation: a duration-weighted, exponentially decaying pitch-class histogram, with all 24 rotations of a major/minor profile pair ranked by Pearson correlation, using the corpus-trained profile values of Albrecht and Shanahan [3] (whose original algorithm pairs them with Euclidean distance). On top of it, two symbolic layers were built: a weighted-evidence layer scoring key-relative chord functions (diatonic membership, dominant and leading-tone function), and a progression layer voting on cadential transition patterns. The resulting three-way hybrid (correlation base plus small functional and progression blend terms) beat the floor on early local-key development runs (+0.096 exact per piece, $p = 0.003$), but the ablations of Section 6 later removed both layers from the selected section-key configuration. Except for the constrained same-tonic mode cue of Section 7, chord-identity features proved useful mainly for the local-key version of the task; the selected section-key configuration is deliberately close to a pure profile model.

The selected detector is a causal HMM over the 24 major and minor keys. It keeps a filtered posterior and updates it by the forward algorithm only; no Viterbi decoding or future context is used. At each event, the update is:

```
for each pitch class p:
  histogram[p] =
    sum over prior events i of:
      duration_i
      * 2^(-age_i / half_life)
      * indicator(p in voicing_i)

for each candidate key:
  profile_score[key] =
    correlation(histogram, key_profile[key])

emission = softmax(profile_score / temperature)
prior =
  transpose(transition) * previous_posterior
posterior = normalize(emission * prior)
```

Multiplication in the last line is elementwise. In words: the profile scores become an event-level likelihood over keys, the previous belief is advanced through the transition model to form a prior, and the two are combined and renormalized into the filtered belief after the current event.

In the selected configuration, `key_profile` is the Albrecht-Shanahan major/minor profile set [3]. Each event's contribution to the histogram is multiplied by its duration, and `half_life = 30 s` means that contribution halves after 30 seconds. The softmax temperature is 0.25, sharpening the profile-correlation scores into a more selective likelihood. The transition matrix assigns 0.9 probability to remaining in the same key; the remaining mass is distributed over other keys with decay by circle-of-fifths signature distance. At each event, the detector reports the posterior's top key and its probability only when the margin between the top two probabilities is at least 0.3; otherwise it abstains.

Two modeling constraints matter for interpreting the results. First, the decayed histogram and the HMM's state persistence are both memories, and evidence entering one is carried forward by the other. An early configuration fed the HMM emission scores that already integrated 30 s of history and expected the transitions to supply key tracking on top; the posterior saturated and lagged behind corrections. The two mechanisms therefore hold distinct jobs: the histogram defines what one observation is, and the transitions supply persistence across hidden keys. Second, the emission memory controls how local each observation is; the next section shows that this dial selects between annotation targets.

6 The timescale finding

Key annotations come in two cultures. Analysts marking local keys (When in Rome) record every tonicization, however brief: a typical piece carries about seven key regions, each only a few chords long. Song keys and key signatures (Isophonics, ASAP) instead name the key of a whole section and absorb short excursions into it. We call these **local-key** and **section-key**

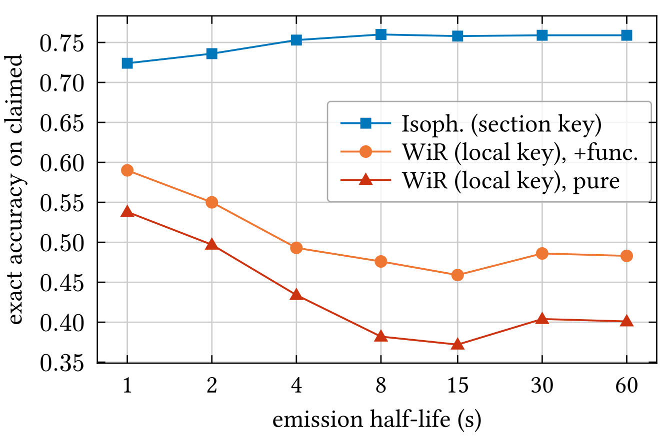

Figure 1: Emission-memory dose-response on the two development annotation scales. The curves run in opposite directions with no interior optimum: the dial selects the reported timescale. Coverage rises with memory on both scales (0.77 to 0.93 on Isophonics); the section-key plateau spans 8-60 s. "Pure" means profile-correlation emissions only; "+func." adds the early chord-function evidence term.

annotations. A detector graded against local keys is rewarded for reacting immediately; one graded against section keys is rewarded for staying put.

Figure 1 sweeps the emission-memory half-life from 1 to 60 seconds on both development corpora, a dose-response design: memory length is the dose, accuracy on claimed events the response. The two label cultures answer in opposite directions. Against local-key labels, accuracy only falls as memory grows, and the detector catches half as many key changes. Against section-key labels, accuracy climbs to a broad plateau from 8 s outward, and longer memory keeps buying coverage and stability (the p90 spurious-switch count falls from 7 to 1). There is no interior optimum to find; the result is the crossover between annotation targets, not a best smoothing value.

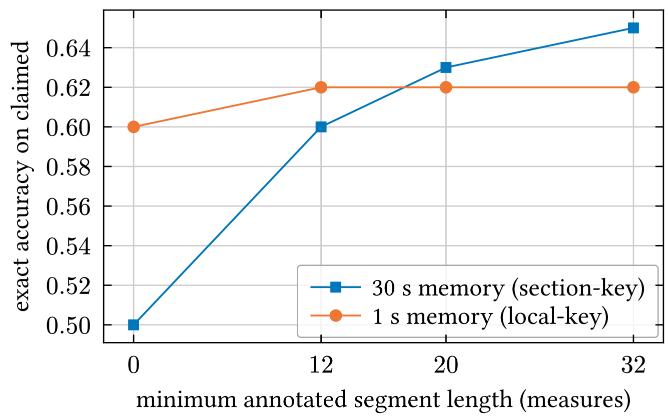

Figure 2: The crossover within a single corpus on identical performed input (ASAP-WiR overlap): filtering to longer analyst key segments, the long-memory configuration climbs and overtakes the short-memory one at 20-measure segments, which stays flat. The same recordings, sliced by label granularity, reorder the two configurations. Accuracy pools claimed events under each segment filter.

Could the crossover instead reflect input noise, or some other difference between the corpora? Three controls say no. First, noise: on the evaluation-only overlap corpus, identical performed recordings scored against analyst local keys still prefer short memory (0.60 vs 0.50 exact on claims, the leftmost points of Figure 2), matching the clean-score result. Second, corpus identity: the crossover reproduces **within** that single corpus when the same claims are re-scored by annotated segment length (Figure 2); the long-memory configuration overtakes as segments lengthen, while the short-memory one stays flat. Third, the crossover holds on held-out data with significance in both directions (Section 9).

The application conclusion is a choice, not a claim that one timescale is universally better: a glanceable key indicator should report the section, so the selected configuration uses 30 s memory and deliberately absorbs brief tonicizations. The research conclusion is that cross-paper accuracy comparisons are under-specified without the label timescale: the same detector moves by tens of points when the annotation scale changes.

Ablations. Four candidate ingredients were tested in every combination, a 16-cell factorial run once per annotation scale: the functional and progression blends of Section 5, duration weighting (a chord's evidence counts in proportion to how long it was held), and recognizer-confidence weighting (evidence is discounted when the recognizer's top two chord readings were nearly tied). Each came back with a clean verdict. The functional blend flips sign with the annotation scale: removing it is a significant paired win on section-key labels, yet on local-key labels it is load-bearing (+0.061 exact, CI95 [+0.019, +0.108], p = 0.010), direct evidence that the functional rules are excursion detectors. The progression blend had earned its place under an earlier architecture, and that value washes out under the HMM. Duration weighting helps on both scales (+0.02 to +0.05 exact) and is kept. Recognizer-confidence weighting is inert in five out of five paired tests, a useful negative because exploiting the recognizer's confidence was a natural hypothesis for this setting. The selected emissions are therefore pure profile correlation. As a bonus nobody engineered, a long-standing behavioral failure vanished with the deleted rules: the functional layer had been hearing every tonic seventh chord in a 12-bar blues as a pull toward the subdominant, and with it gone the blues probes finally read correctly.

Abstention earns its place at this operating point. Sweeping the posterior-margin floor traces the coverage-accuracy curve of Figure 3, which rises monotonically as the detector grows choosier: abstentions are informative, not random.

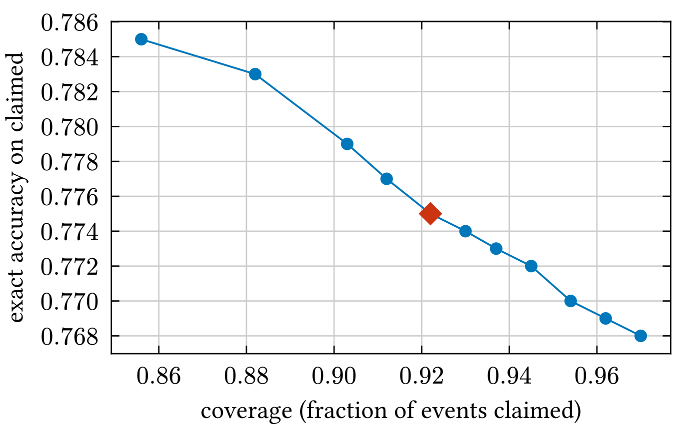

Figure 3: Selective-prediction behavior: the coverage-accuracy curve of the selected configuration on the Isophonics development split, swept over the posterior-margin floor (0 to 0.6). Moving left raises the margin required to speak, so coverage falls; moving up means the remaining claims are more often correct. The marked point is the selected operating point (floor 0.3).

7 Mode disambiguation

On mode-resolved corpora the residual errors sort into MIREX classes: fifth-neighbor confusions, relative-pair confusions (shared signature), and parallel-pair confusions (shared tonic). Mode errors, the most user-visible class since an indicator displays major or minor, were about 10% of claims on both annotation scales once the timescale trade was filtered out.

The design constraint that works is mass preservation within a musically related pair: if a cue only redistributes evidence between two hypotheses that share an invariant, it can help choose between them while leaving every other key untouched.

Parallel pairs (selected). Parallel keys share the tonic pitch class, and practical minor borrows the raised sixth and seventh, so the discriminating evidence collapses to roughly one pitch class, easily outvoted in a 12-dimensional correlation. Symbolic observations change that: when the played chord is rooted on a candidate tonic and its quality is one a tonic chord can carry (plainly major or minor), that quality is the most direct mode cue available. The cue is applied as a log-odds tilt within the parallel pair after the emission softmax, then rescaled so the pair's total emission is unchanged. By construction it cannot move tonic evidence at all, which rules out the failure mode of two less constrained ancestors, an unconditional bonus for tonic chords and the functional rules of Section 5, both of which could fight modulation tracking. Strength 2 was selected after a 0.25-to-4 sweep with a broad 2-to-4 plateau: paired exact wins on both annotation scales (Isophonics +0.016, p = 0.030; When in Rome +0.030, p = 0.029), parallel confusion roughly halved everywhere measured (4% to 2% of claims on Isophonics), and no stability cost on the section-key development set.

Relative pairs (measured negative). The same pattern generalizes with the key signature as the conserved quantity. But here the evidence is structurally weaker. In a parallel pair, hearing the rival's tonic chord is rare mode mixture; in a relative pair it is ordinary diatonic harmony (A minor's tonic triad is just the vi chord of C major), so isolated chord quality fires constantly for the wrong twin. Two sharpened cues were built and swept: a bass-gated tilt (fires only when the chord's root is also its bass) and a cadential-bigram tilt (a dominant-quality chord resolving down a fifth onto a home-quality chord). The cadence cue is inert, echoing the progression ablation. The bass cue works mechanically (relative confusion 6% to 4% of claims) but misses paired significance on the primary scale (+0.007, p = 0.055), trends negative on the other (-0.009, p = 0.056), and doubles spurious p90; unlike the parallel tilt's broad plateau, it turns harmful at strength 2, the signature of a weak signal. Not adopted; retained in code as a measured negative.

Fifth pairs (not adopted). Fifth neighbors conserve neither tonic nor signature, and every key has two of them, so there is no pair to redistribute within. Discriminating a key from its dominant **is** the modulation problem, and any chord-quality nudge between fifth neighbors recreates the unconditional tonic bonus rejected above. Empirically the residual fifth errors also lack an exploitable shape, splitting nearly evenly between the dominant and subdominant sides (57/43 and 56/44 on the two mode-resolved corpora).

One pattern explains all three outcomes: a tilt is worth what its invariant is worth. Parallel keys share a tonic, and the discriminating evidence is rare and decisive, so the cue was adopted. Relative keys share only a signature, and the discriminating evidence is everyday harmony, so the cue was too weak. Fifth neighbors share nothing, so there is no pair to tilt within at all.

8 Adaptive temporal models

Explicit-duration modeling (HSMM). The HMM's fixed self-transition implies geometric key-dwell times: at every event the key survives with the same probability. An HSMM (hidden semi-Markov model) would replace that with a richer dwell-time family. Before building one we measured whether the system even responds to the parameter an HSMM would refine: sweeping the self-transition from 0.7 to 0.98 (mean dwell roughly 3 to 50 events) moves every metric along the coverage-accuracy curve rather than off it (Isophonics exact within 0.008, spurious p90 pinned at 1). The probe does not prove explicit-duration modeling useless for key estimation; it shows there is little headroom at the selected operating point.

Bayesian online changepoint detection, built and measured. The dwell probe addresses the changepoint prior (constant-hazard BOCPD implies the same geometric dwell) but not BOCPD's distinct contribution, the adaptive evidence window: instead of decaying evidence through a fixed 30 s half-life, BOCPD infers where the current section started and pools evidence back to that point. We built the constant-hazard detector for the 24-key space [10] with emission conventions identical to the selected HMM (including the mode tilt), so any difference is attributable to the window (Table 2).

config	cov	exact	modulations	spur med/p90
HMM, selected	0.92	0.775	94/192	0/1
BOCPD h=1/200	0.88	0.715	161/192	5/14
BOCPD T=0.5	0.89	0.765	135/192	2/7
BOCPD T=1.0	0.90	0.769	115/192	0/4

Table 2: BOCPD versus the selected HMM on the section-key development set (Isophonics dev). Here `h` is the constant changepoint hazard and `T` is emission temperature. The adaptive window improves modulation detection but no tuning recovers the selected stability point.

The adaptive window delivers its promise: modulation matching jumps by 70%, because evidence resets at inferred section starts. But the same reactivity breaks section-key stability, and no tuning recovers the selected operating point: softening the per-event evidence walks BOCPD back along the frontier without reaching the HMM (best cell: exact wash, p = 0.51, spurious p90 still 4 versus 1). One finding is worth keeping: at matched reactivity BOCPD dominates the HMM's fast settings (0.765 exact at 135 modulations versus 0.736 at 141 for the HMM at a 2 s half-life), so adaptive windowing has real value for local-key tracking; it simply was not selected for the section-key task studied here. The false alarms are within-section harmonic movement, exactly where BOCPD's independent-given-key assumption is weakest. Autoregressive extensions target the analogous within-regime dependence in continuous series [11], but those Gaussian remedies have no direct analog for categorical chord emissions, and our domain's substitute, the progression layer, is measured inert.

9 Held-out evaluation

The test splits were evaluated exactly once, with the result set declared before any run: the selected configuration on all three splits; the local-key reference configuration on the two mode-resolved splits; the music21 [19] baselines on Isophonics, with paired and matched-coverage comparisons; and a descriptive mode-confusion breakdown. The complete artifacts are committed with the project.

split	pieces	cov	exact	mods	spur
Isophonics	41	0.88	0.732	10/22	0/3
WiR	18	0.81	0.587	39/115	0/1
ASAP	10	0.83	0.683	-	0/0

Table 3: The selected configuration on the three held-out splits. `mods` is matched annotated modulations; `spur` is median/p90 spurious switches. WiR is When in Rome; ASAP is scored with acceptable keys because labels are mode-unknown key signatures.

Generalization is clean (Table 3): Isophonics dips modestly from development (0.775 to 0.732), When in Rome comes in far above it (0.434 to 0.587, easier held-out pieces), and stability holds everywhere; the differences run in both directions, not the uniform degradation a tuned-to-development configuration would show.

The held-out crossover is significant in both directions. The short-memory local-key configuration is better when judged against local-key labels (0.649 vs 0.587 exact, paired +0.062, CI95 [+0.004, +0.121], p = 0.047). The selected section-key configuration is better when judged against section-key labels (0.732 vs 0.556, paired +0.175, CI95 [+0.040, +0.315], p = 0.039), while also avoiding the local-key configuration's spurious switches (0/3 vs 5/11 med/p90).

External comparison. Table 4 compares against music21's offline whole-piece analyzers on the held-out songs.

system	coverage	exact	MIREX
this work (causal, abstaining)	0.88	0.732	0.782
Temperley-Kostka-Payne [4]	1.00	0.637	0.740
Krumhansl-Schmuckler [1]	1.00	0.624	0.726
Aarden-Essen [20]	1.00	0.558	0.690

Table 4: Held-out Isophonics test split: the causal, abstaining detector against offline whole-piece baselines.

The causal detector's point estimate leads every offline analyzer, including at matched coverage (Krumhansl-Schmuckler restricted to our claimed events: 0.63). The paired test is a wash (+0.108, CI95 [−0.008, +0.228], p = 0.25), so the defensible claim is **at least parity under a more constrained setting**: the baselines read each entire song before answering; this detector never sees the future, abstains on the ambiguous 12%, and reports posterior probabilities and streaming stability metrics offline systems do not have.

The descriptive confusion mix is: exact 72%, fifth 7%, relative 5%, parallel 2%, other 14%. This pooled exact rate differs from the per-piece mean in Table 3; the parallel row matches the mode tilt's development effect exactly.

Posterior calibration. Raw filtered posteriors are overconfident: on claimed exact-labeled development events, the mean top posterior is 0.929 against 0.772 exact accuracy (expected calibration error, ECE, 0.157). Overconfidence does not affect ranking, abstention, or any paired accuracy claim, but it matters the moment the probability is displayed. Post-hoc temperature scaling [21], fit once on the development split (T = 1.55, negative log-likelihood argmin), reduces held-out claimed-event ECE from 0.192 to 0.041. Because the transform is monotone and display-only, the held-out claims are byte-identical to the committed one-shot artifacts, which simultaneously verifies that the calibration pass took nothing from test data.

10 Limitations

The label timescale is a construct of the annotation culture, and our central finding is precisely that results are relative to it; we mitigate by reporting both annotation scales, but neither scale is "the truth."

Residual error contains genuine ambiguity (relative twins share every pitch class; mode-mixture passages and dominant prolongations can reasonably be notated different ways), so ceilings are below 1.0 and unknown.

The recognizer sits inside the loop: fixtures embed its rankings, so recognizer changes can change detector results and require regenerating fixtures before comparison.

Corpus licensing constrains redistribution (two corpora are noncommercial- or research-gated; we commit derived facts, splits, and evaluation artifacts, not the gated fixtures).

Test splits are small (10 to 41 pieces), which the paired statistics respect but cannot cure. Stronger neural local-key systems are offline and score-hungry; we compare only against baselines runnable on our fixtures. In particular, justkeydding [6], the closest published analogue, did not build reproducibly in our environment, so no comparison claim is made against it in either direction.

Calibration is task-relative, exactly like accuracy: the temperature-scaled display of Section 9 is calibrated to section-key correctness on pop material, and against local-key labels the same display remains overconfident.

11 Conclusion

This paper asked whether a live application can tell a musician what key they are in, from chord-recognition output alone, without ever seeing ahead. The answer is yes, and with less machinery than expected. Under a frozen protocol, paired statistics, and a held-out evaluation performed exactly once, a causal HMM over profile-correlation evidence, plus one constrained major-versus-minor cue, reports section keys at least on par with offline baselines that read the whole song first. The route there was largely simplification guided by measurement: functional and progression rules help only when the target is local-key tracking, recognizer-confidence weighting is inert, and adaptive temporal models trade stability for reactivity. The lasting methodological lesson is that key-detection accuracy is label-timescale dependent. The same detector looks better or worse depending on whether the reference labels mark brief tonicizations or section keys, so the annotation scale is part of the task definition, not a detail of the dataset.

Two follow-ups are natural. Local-key tracking deserves treatment as a task in its own right rather than as an ablation of the section-key detector: short memory, the functional blend, and adaptive windowing all have measured value in that regime. And the fixture design's deliberately neutral analysis context leaves open a closed-loop question, whether feeding the detector's tonality belief back into chord recognition improves the chord identities enough to change key tracking, which requires evaluating the coupled system rather than a fixed event stream.

Competing Interests

The author has no competing interests to declare.

Reproducibility: protocol, logs, corpus pins, splits, harness, and held-out evaluation artifacts live under `research/whatkey/` in the WhatChord repository. Every number traces to a dated log entry and a report with its generating command.
https://github.com/EarthmanMuons/whatchord

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